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MPTRAC: A high-performance Lagrangian transport model for atmospheric air parcel dispersion

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Summary

The Massive-Parallel Trajectory Calculations (MPTRAC) Lagrangian transport model (Hoffmann et al., 2016, 2022) simulates the movement, dispersion, and chemical transformation of aerosol particles and trace gases in the atmosphere. Unlike grid-based models, MPTRAC tracks individual air parcels, offering high-resolution insights into transport, mixing, chemistry, and deposition. Trajectory calculations are driven by meteorological fields from data assimilation systems and forecast models. MPTRAC has been applied in various contexts, especially for tracking volcanic plumes in the troposphere and stratosphere. Optimized for high-performance computing (HPC) and graphics processing unit (GPU) systems, MPTRAC efficiently supports large-scale, high-resolution simulations, making it a powerful tool in atmospheric research.

Statement of need

Accurate atmospheric transport modeling is essential for applications such as air quality assessments, climate studies, and public health protection. Traditional Eulerian models often struggle with fine-scale resolution, limiting their ability to capture complex transport and mixing processes. Lagrangian models, such as MPTRAC and others (Jones et al., 2007; Lin et al., 2003; McKenna et al., 2002; Pisso et al., 2019; Stein et al., 2015; Stohl et al., 2005), address this limitation by tracking individual air parcels, offering fine-grained representations of atmospheric processes.

MPTRAC supports both research and operational applications requiring high-precision transport simulations. It models long-range transport and chemical transformations, enabling pollution studies, emission tracking, and environmental impact forecasting. MPTRAC runs efficiently on HPC and GPU systems, supporting fast, large-scale simulations that facilitate real-time decision-making and scientific analysis.

Features

MPTRAC computes air parcel trajectories using wind and vertical velocity fields from global reanalysis or forecast data (Gelaro et al., 2017; Hersbach et al., 2020). It accounts for eddy diffusion and subgrid-scale winds via the Langevin equation. An inter-parcel exchange

module represents air mixing. Additionally, MPTRAC simulates convection, sedimentation, radioactive decay, gas and aqueous phase chemistry, and wet and dry deposition. Meteorological pre-processing routines compute boundary layer heights, CAPE, geopotential heights, and tropopause data. Figure 1 shows MPTRAC's geophysical and chemical modules and core software components.

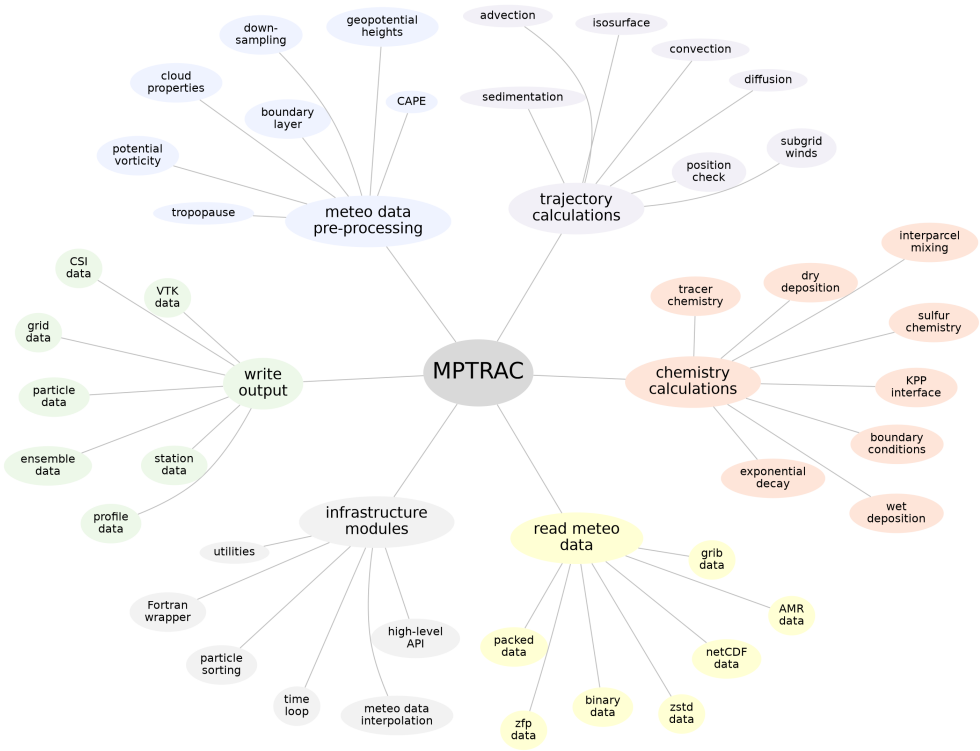

Figure 1: Geophysical and chemical modules and main software components of the MPTRAC Lagrangian transport model. Image adapted from Hoffmann et al. (2024).

Optimized for computational efficiency, MPTRAC features an MPI-OpenMP-OpenACC hybrid parallelization for scalable deployment on workstations, HPC systems, and GPU platforms (Hoffmann et al., 2022, 2024; Liu et al., 2020). It supports multiple output formats for diverse data types, enabling visualization and analysis using Gnuplot, ParaView, or Python. MPTRAC is open-source, distributed under the GNU GPL v3 license. Code and documentation are available in a GitHub repository, with software releases archived on Zenodo.

Applications

MPTRAC simulates the transport and dispersion of atmospheric constituents from both natural and anthropogenic sources. For volcanic eruptions, it helps estimate emissions and track the spread of volcanic ash and sulfate aerosols, which impact air traffic, climate, and ecosystems (Cai et al., 2022; Heng et al., 2016; Mishra et al., 2022; Wu et al., 2017, 2018). Similarly, MPTRAC models the dispersion of carbon dioxide from wildfires, including vertical ascent and widespread transport into the upper atmosphere (Liao et al., 2024). Further studies have examined the long-range transport of aerosol particles and trace gases in the upper troposphere and lower stratosphere (Clemens et al., 2024; Smoydzin & Hoor, 2022; Wu et al., 2023).

Figure 2 illustrates MPTRAC's use in studying convective transport of air from the boundary layer into the free troposphere (Hoffmann et al., 2023). The simulation tracks air parcels lifted by updrafts from tropical storms and mid-latitude systems, revealing how boundary layer air

and pollutants spread aloft, improving understanding of atmospheric circulation and pollutant distribution at regional and global scales.

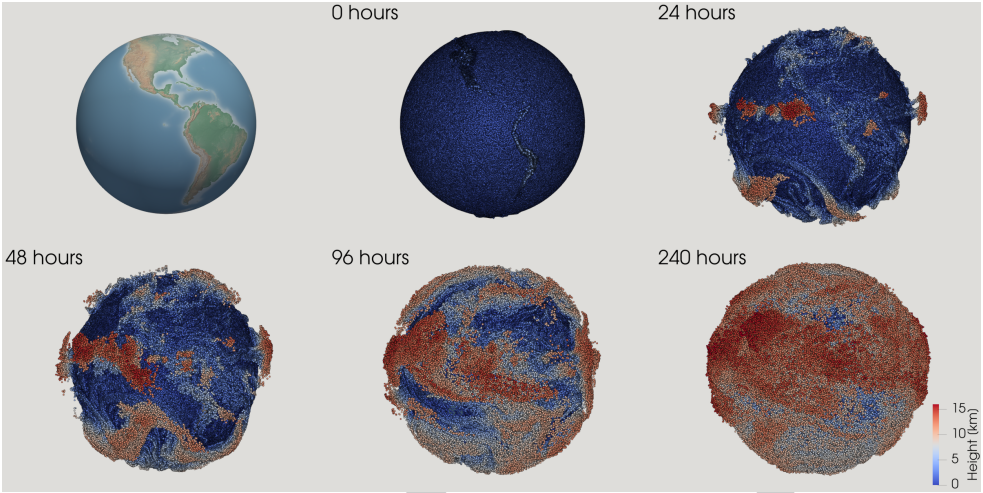

Figure 2: Lagrangian transport simulation of convective transport from the boundary layer into the free troposphere. One million trajectories are initialized near the surface on 1 July 2017, 00:00 UTC, and tracked over 10 days using ERA5 reanalysis data. Color coding indicates the geopotential height of the air parcels. Video available via Zenodo (Hoffmann, 2025).

Evolution and Future Directions

MPTRAC has evolved since 2013, originally designed for HPC applications. Initially using OpenMP for multi-core CPUs, it later incorporated MPI for large-scale ensemble simulations. In 2019, OpenACC offloading boosted performance by porting all geophysical modules to NVIDIA GPUs. In 2023, we explored GPU offloading using OpenMP for AMD GPUs, broadening portability. MPTRAC is now ready for simulations on the Exascale JUPITER system.

Recent technical efforts have improved documentation and usability. Written in C, MPTRAC includes a Fortran wrapper and high-level API, enabling seamless integration of Lagrangian transport simulations into other models. Continuous testing via GitHub Actions and multiple HPC systems, including the JUPITER Exascale Development Instrument (JEDI), ensures robust performance and reliability.

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DRAFT

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DRAFT

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